ANALYSIS REPORT

Wave 5.2R Full-Candidate TE Curve Verification Decision

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Decision Summary

The bounded forward-only `TE Curve Verification Pipeline` refresh evaluated all `98` eligible candidates on the same `97` held-out `polished_dataset + setpoints + Fw` curves.

The official direction-parallel decision is

- `wave52r_stage9_k01` becomes the verified offline temporal-lane leader;
- `wave52r_stage5_h08_seed_314159` becomes the verified offline non-temporal balanced leader;
- `wave52r_stage12_f01` remains visible as the strongest centered-shape specialist;
- `wave52r_stage12_s01` remains visible as the strongest harmonic-amplitude and phase specialist in the diagnostic shortlist;
- the accepted periodic GRU and periodic harmonic MLP remain the deployable reference models until the new lane leaders pass export, parity, runtime, and PLC-facing acceptance checks. This is not a backward or global decision. No `Bw` or `global` result is inferred from forward-only checkpoints.

Evaluation Scope

Item	Value
Dataset	`polished_dataset`
Input contract	setpoints
Surface	`Fw`
Held-out curves	97
Matrix candidates	98
Temporal candidates	18
Non-temporal candidates	79
Analytical anchors	1
Full-payload diagnostic shortlist	10
Exact duplicate prediction groups	3

The complete inventory also records 27 calibration-only, replay, or synthetic artifacts that are not distinct real-data TE predictors.

Curve-First Leaders

Temporal Lane

Axis	Leader	Evidence
Raw error	`wave52r_stage9_k01`	Mean MPE `2.716282%` ; mean curve MAE `0.001374 deg`
Centered shape	`wave52r_stage12_f01`	Centered curve MAE `0.001147 deg`
Offset	`wave52r_stage9_k01`	Mean absolute curve-mean error `0.000496 deg`
Harmonic amplitude	`wave52r_stage12_s01`	Mean amplitude error `5.702734%`
Harmonic phase	`wave52r_stage12_s01`	Mean phase error `3.166801 deg`
Robust P95	`wave52r_stage12_f01`	P95 MPE `6.554012%`
Balanced recommendation	`wave52r_stage9_k01`	Best raw, offset, and full diagnostic score

`K01` improves the accepted periodic GRU reference from `3.278427%` to `2.716282%` mean MPE, a relative reduction of approximately `17.15%` . Its centered curve MAE also improves from `0.001382 deg` to `0.001230 deg` , and its mean harmonic amplitude and phase errors are materially lower.

`F01` has the best centered-shape and P95 evidence, but its worst-condition MPE is `14.230071%` , above the `12.059574%` recorded by `K01` . It therefore remains a specialist rather than the balanced temporal recommendation.

Non-Temporal Lane

Axis	Leader	Evidence
Raw error	`wave52r_stage10_r00`	Mean MPE `3.422123%` ; mean curve MAE `0.001658 deg`
Robust P95	`wave52r_stage10_s01`	P95 MPE `6.751695%` ; worst MPE `11.301458%`
Centered shape in shortlist	`wave52r_stage5_h04_seed_314159`	Centered curve MAE `0.001357 deg`
Harmonic and phase balance	`wave52r_stage5_h08_seed_314159`	Amplitude `10.375270%` ; phase `4.087762 deg`
Balanced recommendation	`wave52r_stage5_h08_seed_314159`	Best multi-index compromise without Stage 10 harmonic collapse

The accepted periodic harmonic MLP remains slightly better than `H08` by raw mean MPE, `3.438600%` versus `3.483289%` . However, `H08` improves centered curve MAE from `0.001390 deg` to `0.001364 deg` , harmonic amplitude error from `14.751961%` to `10.375270%` , and phase error from `10.572248 deg` to `4.087762 deg` .

Stage 10 `R00` and `S01` are raw and robustness leaders, but their mean harmonic amplitude errors exceed `34%` and their mean phase errors exceed `31 deg` . The multi-index policy therefore vetoes them as balanced recommendations despite their attractive scalar error.

H04 remains an important structured grey-box result, but it is not the non-temporal lane leader in this expanded comparison. H08 has better raw error and materially stronger harmonic and phase fidelity.

Cross-Lane Interpretation

The temporal road remains strongest on this forward surface. `K01` leads `H08` in raw error, centered shape, offset, harmonic amplitude, harmonic phase, and the combined curve-payload diagnostic score.

The two-road strategy remains useful

- advance `K01` as the primary temporal export and runtime-validation target;
- advance `H08` as the primary inspectable non-temporal research target;
- retain `F01` and `S01` as temporal specialists for future multi-head or constrained optimization work;
- retain the accepted GRU and harmonic MLP as operational fallbacks until the new leaders pass deployment acceptance.

Deployment Decision

The offline lane-leader status changes, but the deployment baseline does not change automatically.

`K01` and `H08` currently rely on reproducible training or prediction artifacts and passed the forward curve-verification checks. Promotion into the accepted deployable model set requires:

1. a frozen standalone export;
2. Python-to-export numerical parity;
3. bounded runtime and memory evidence;
4. causal full-curve replay without hidden future information;
5. TwinCAT or PLC-facing integration checks where applicable.

   Until those gates pass, the periodic GRU and periodic harmonic MLP remain the accepted deployable references.

Evidence

- matrix summary: `output/validation_checks/track2_reference_comparison/2026-07-30-10-45-46__wave52r_full_candidate_parallel_temporal_non_temporal_wave52r_full_candidate_parallel_temporal_non_temporal/validation_summary.yaml` ;
- curve-first ranking: `output/validation_checks/wave52r_full_candidate_track2_curve_first_reranking/2026-07-30-11-09-17__track2b_curve_first_reranking/` ;
- curve-payload diagnostics: `output/validation_checks/wave52r_full_candidate_track2_curve_payload_diagnostics/2026-07-30-11-00-18__track2c_curve_payload_diagnostics/` ;
- candidate collage: `doc/reports/analysis/te_curve_verification_pipeline/02_visual_reports/wave52r_full_candidate_best_model_collage_report/[2026-07-30]/` ;
- finalist overlay: `doc/reports/analysis/te_curve_verification_pipeline/02_visual_reports/wave52r_full_candidate_multi_model_curve_comparison_report/[2026-07-30]/` .

Final Status

The Wave 5.2R full-candidate forward verification is closed.

- Temporal offline lane leader: `wave52r_stage9_k01`.
- Non-temporal offline balanced leader: `wave52r_stage5_h08_seed_314159`.
- Accepted deployment baselines: unchanged pending export and runtime gates.
- Backward and global surfaces: unchanged and outside this verification scope.